

Reading a ratio

A ratio compares two quantities by dividing one by the other. Writing $a : b$ for the comparison of a with b , the ratio is the single number

$$r = \frac{a}{b}, \quad b \neq 0.$$

A recipe using 200 g of flour for 100 ml of milk has a flour-to-milk ratio of 2, whatever quantity is actually made.

Equivalent ratios

Multiplying both quantities by the same non-zero number leaves the ratio unchanged:

$$\frac{ka}{kb} = \frac{a}{b}, \quad k \neq 0.$$

So $200 : 100$, $400 : 200$ and $2 : 1$ are the same ratio written three ways. This is the property every later use depends on.

Ratios and fractions

A ratio compares two quantities of the same kind, and the result has no unit. Dividing quantities of different kinds gives a rate instead, such as the distance-over-time of an average speed.

$$\begin{matrix} a & b \\ \hline ka & kb \end{matrix}$$

Both rows above describe the same ratio, which is what makes the table a legitimate way to record one.

Scaling a recipe

Rescaling a recipe means changing every quantity while keeping every ratio the same. One scale factor does the whole job.

To go from a recipe serving n people to one serving m , the scale factor is

$$k = \frac{m}{n},$$

and every quantity q becomes kq .

Worked example

A recipe for 4 people uses 200 g of flour and 100 ml of milk. For 6 people, $k = 6/4 = 1.5$, giving 300 g of flour and 150 ml of milk.

The flour-to-milk ratio is still 2, as it must be: both quantities were multiplied by the same k .

The usual mistake

Adding a fixed amount to each quantity instead of multiplying does not preserve the ratio. Adding 100 to both gives $300 : 200$, a ratio of 1.5 rather than 2, so the result is a different recipe.

A quick check is to rescale, then recompute one ratio. If it moved, the scaling was additive somewhere.